

TAMAJYOTI DHAR

Data & AI Engineer

tamajyoti.dhar@gmail.com | +91-9008054298 | linkedin.com/in/tamajyoti-dhar-6494022b

PROFESSIONAL SUMMARY

Data & AI Engineer with 14+ years architecting production-scale streaming and lakehouse platforms and building AI-native systems agentic workflows, RAG/retrieval, and real-time ML across fintech, gaming, and advertising technology. Proven track record owning platform strategy end to end (ingestion, streaming, warehousing, observability) while leading the layer of AI and agentic tooling built on top of it, from real-time recommenders to LLM-orchestrated multi-agent systems to data-access products (MCP) serving quant and analytics teams.

CORE COMPETENCIES

Data Platforms & Streaming: Kafka, Spark (Structured Streaming/PySpark), ClickHouse, Snowflake, Redshift, Apache Iceberg, Trino, PostgreSQL, Elasticsearch/OpenSearch, Event-Driven Architecture, Lakehouse Design

AI & LLM Systems: LangGraph, LangChain, RAG Pipelines, Multi-Agent Orchestration, MCP (Model Context Protocol), Semantic & Hybrid Retrieval, Prompt Engineering, PyTorch, Recommendation Systems

Orchestration & Infra: dbt, Airflow, Dagster, AWS (S3/EC2/EMR/Glue/Lambda), Kubernetes, Docker, Terraform, CI/CD, MLflow

Observability & Governance: Prometheus, Grafana, Data Contracts, Schema Governance, SLA Monitoring, RBAC, Data Lineage

PROFESSIONAL EXPERIENCE

Independent Data & AI Consultant

Jan 2026 – Present

Engaged by clients on a contract basis to lead data platform architecture and AI/agentic system delivery.

Client: Khawk (Quantitative Trading Platform) — Senior Data Engineer

May 2026 – Present

Engaged to build the real-time market data infrastructure powering trading strategy development, backtesting, and stress testing.

- Architected an end-to-end Kafka-to-ClickHouse pipeline achieving sub-second latency for streaming trades data, supporting real-time consumption by downstream strategy and analytics systems.
- Deployed and scaled 40+ Kafka topic consumers on Kubernetes, managing partitioning, throughput, and fault tolerance for high-volume trades data ingestion.
- Own ClickHouse infrastructure end to end from schema design, query performance, and operational reliability to serving as the system of record for trades data at scale.
- Built and deployed an MCP (Model Context Protocol) server exposing trades data through internal and external APIs, enabling quant and strategy teams to programmatically access data for backtesting and stress-testing workflows.

Client: Mint.ai (Advertising Technology Platform) — Senior Data & AI Platform Consultant

Jan 2026 – Apr 2026

Engaged to modernize a large-scale advertising resource management platform and build an AI-powered creative discovery and campaign planning layer on top of it, supporting 40,000+ monthly reporting workloads.

- Re-architected Snowflake ingestion frameworks improving schema governance and workload isolation along with a memory-aware distributed ingestion engine that eliminated recurring Kubernetes OOM failures, cutting ingestion runtime from 6 hours to under 2 hours.
- Established platform-wide observability using Prometheus and Grafana with automated data quality validation and freshness checks reducing time-to-detection from hours to minutes.
- Built a LangGraph-based multimodal campaign planning workflow with structured tool routing across four specialized agents (asset search, insight retrieval, budget allocation, report generation), reducing planning turnaround from ~2 days to under 2 hours.
- Designed a retrieval pipeline across historical ad performance and creative asset datasets using dense search and sparse indexing, reducing manual search effort by ~65%.
- Partnered with cross functional teams to design data models for downstream AI/ML models using dbt pipelines.

Senior Data & AI Engineer | Drest, London (Remote)

Nov 2022 – Jan 2026

Primary architect and owner of Drest's end-to-end data platform for a live fashion gaming app, evolving it from an early-stage warehouse into a near real-time cloud-native lakehouse — then leading the build of real-time personalization and AI systems on top of it as the platform matured.

- Established the foundational data platform architecture using AWS, Redshift, Airflow, and dbt, enabling sustainable growth from 1.2M to ~250M daily events.
- Led migration from a Redshift-centric architecture to a cloud-native lakehouse on Apache Iceberg and Trino, reducing infrastructure costs and enabling independent compute/storage scaling.
- Introduced an event-driven platform powered by Kafka and Spark Structured Streaming, reducing analytical latency from overnight batch to near real-time.
- Designed a Kafka-to-ClickHouse real-time analytics platform supporting traffic spikes of 10–15x normal load during high-concurrency events, enabling sub-minute reporting latency.
- Established platform-wide engineering standards like schema governance, data contracts, observability, SLA monitoring maintaining 99.9% pipeline reliability.
- Designed and deployed a near real-time recommendation system using feature engineering and adaptive retrieval to personalize player experiences across millions of daily gameplay interactions.
- Built a hybrid retrieval pipeline (ANN search, metadata recall, RRF candidate merging) with a LangGraph-orchestrated agent dynamically routing between warm, cold-start, and trending-boost strategies keeping the full recommendation path under 30ms p50.
- Built and trained a two-tower PyTorch model with in-batch negative sampling and a LambdaRank reranker for listwise scoring, served via SageMaker endpoints with weekly automated retraining triggered by NDCG drift detection.
- Evolved recommendation logic into a three-agent AI system (profile, retrieval, ranking) with structured tool routing and LLM-free hot paths, with full observability over retrieval quality and agent execution.

Senior Data Engineer | Superside (Global Creative Ops Platform)

Oct 2021 – Nov 2022

Senior member of the analytics platform modernization initiative, leading migration from a legacy AWS Glue/Druid stack to a centralized Snowflake, dbt, and Airflow platform.

- Led migration from AWS Glue, Druid, and ad-hoc SQL pipelines to Snowflake, dbt, and Airflow, consolidating business logic into version-controlled, testable transformation workflows.
- Designed migration validation and reconciliation frameworks enabling accurate historical backfills with zero data discrepancies.
- Led Snowflake optimization (clustering, warehouse sizing, workload management), reducing query runtimes by 40–60% across core analytics workloads.
- Established data governance foundations with environment isolation, RBAC, validated migration workflows enabling safe concurrent development.

ML / Data Engineer | ComplyAdvantage, London

Jul 2019 – Oct 2021

- Built distributed multilingual document ingestion (Kafka, Spark Structured Streaming) and Elasticsearch search infrastructure across 10M+ risk entities for a global AML/risk intelligence platform.
- Trained multiclass risk-document classifiers and a fine-tuned BERT+CRF extraction pipeline, with a three-stage entity resolution system (blocking, XGBoost matching, connected-components resolution) at corpus scale.

Earlier Career

2012 – Jul 2019

Data Engineer, BlueJeans (Verizon) — fraud/anomaly detection and network-load forecasting pipelines. Lead Analytics Engineer, Belong — built analytics function from scratch (ETL, KPI frameworks, churn prediction, A/B testing); led a team of 4. EY / HSBC / Infosys — AML model development and regulatory reporting across banking and consulting engagements.

EDUCATION

PGDM, Operations — Welingkar Institute of Management, Bangalore (2010 – 2012)

B.Tech, Electronics & Telecommunications — NERIST (2006 – 2010)